Symbolic signed-graph reduction for pencil relations

Mathematical implementation and development checks

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The existing pencil cohort has 543 verified candidate targets out of 1,260. This packet adds a native algebraic component, with no new cohort or performance measurement. Its candidate/rho cost ratio is unknown.

Algebraic interface

An input row is $\sum a_i X_i + \sum b_j S_j = 0$ over a prime field. At most two large-variable terms survive canonicalization; the S_j remain symbolic. The forest stores $X_i = s_i X_{\mathrm{parent}(i)} + u_i(S)$ and uses union by size. Root substitution leaves zero, one or two free roots. Zero emits a constraint, one anchors a root, and two join components by an invertible pivot. An unbalanced signed cycle anchors its root; it does not immediately yield an additional relation involving only small variables.

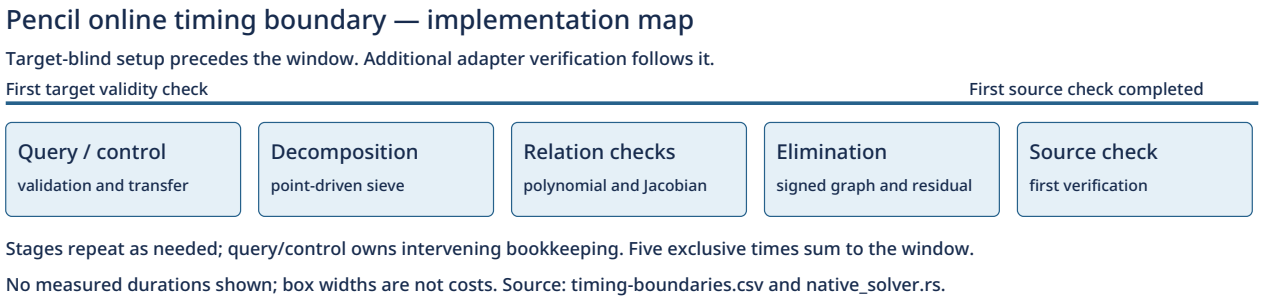
Recorded development checks

The Rust suite compares the emitted small-row space with independent dense row reduction after every prefix: 40 prefixes, 128 deterministic streams and five prime fields (2, 3, 5, 7 and 101), totaling 25,600 checks. Additional cases cover signed cycles, duplicate terms, rejected inputs, composites and a prime near the 64-bit limit. The test log is retained. The four Lean declarations cover pivot equivalence, substitution, balanced cycles and unbalanced cycles. Their direct compiler result and axiom report are retained separately. They do not formalize the full forest invariant or prove correspondence with Rust. The retained residual basis is now also compared with the independent reference at all 25,600 prefixes. Fourteen default library tests pass. Explicit working and retained coefficient caps leave the basis unchanged on rejection; callers must stop or retain that rejected row. Original residual diagnostics count failed attempts; the separate semantic and allocation observations described below extend their coverage. These are coefficient limits, not process-memory guarantees or a complete cost model. The composed adapter binds both stages to one field and fixed column partition, rejects three-large-variable rows explicitly, and retains an unincorporated constraint on terminal cap. Its row space is also checked at all 25,600 prefixes. The optional producer now connects this adapter to the existing geometric checks and matched control. Producer tests check signed coefficient parity, rejection of constant-only inconsistencies and phase attribution of symbolic events. Setup is a separate recorded phase. Semantic arithmetic counts include inversion multiplications and distinguish primality powers from inversions. Direct pivot lookup replaces full basis scans; no performance measurement is inferred. Residual diagnostics overlap this ledger and must not be double-charged. The producer has not run a cohort. Its new schemas explicitly mark costs incomplete and ratios null; a frozen partition and full accounting remain open. Allocation observations now cover cold setup, phases, the whole candidate envelope and rho. A grow/shrink regression checks payload preservation and requested-layout accounting. These bytes are not RSS or internal copying; rho's absolute peak includes still-live candidate and shared input state. The outer envelope includes between-phase work without losing peaks when inner phases reset. No runtime or memory comparison has been measured. Sparse maps now record API calls, ordered comparisons, yielded entries and cloned entries.

Exact iterator size hints and numeric ordering are preserved; JSON serialization is unmetered. These are semantic events, not all CPU work. Counts depend on the recorded standard-library version. The current suite has 14 default library tests and 28 passing producer-feature tests, including eight solver tests. One shared-clock test is ignored. The producer now verifies field, native arithmetic and symbolic phase sums, and retains interval overhead separately. Scalar inverse instrumentation was asymmetric: pencil counts inner multiplications; the reference records one inverse. A native normalization removes pencil inverse multiplications for a common atomic scalar-I model, tested against both actual inverse implementations. The model remains a draft with unfrozen weights; no historical ratio is revised. The shared per-target API now separates arithmetic from archive input, known-answer checks and rho pairing. Version 2 repeats cold setup per target and charges target validity. Its allocation baseline changes because setup temporaries drop before rho. Missing allocation observers produce nulls. The root ecbench method is registered with an explicit serialization-only kernel dependency. Root validation and prerequisite rechecks are recorded below; cohort execution remains open.

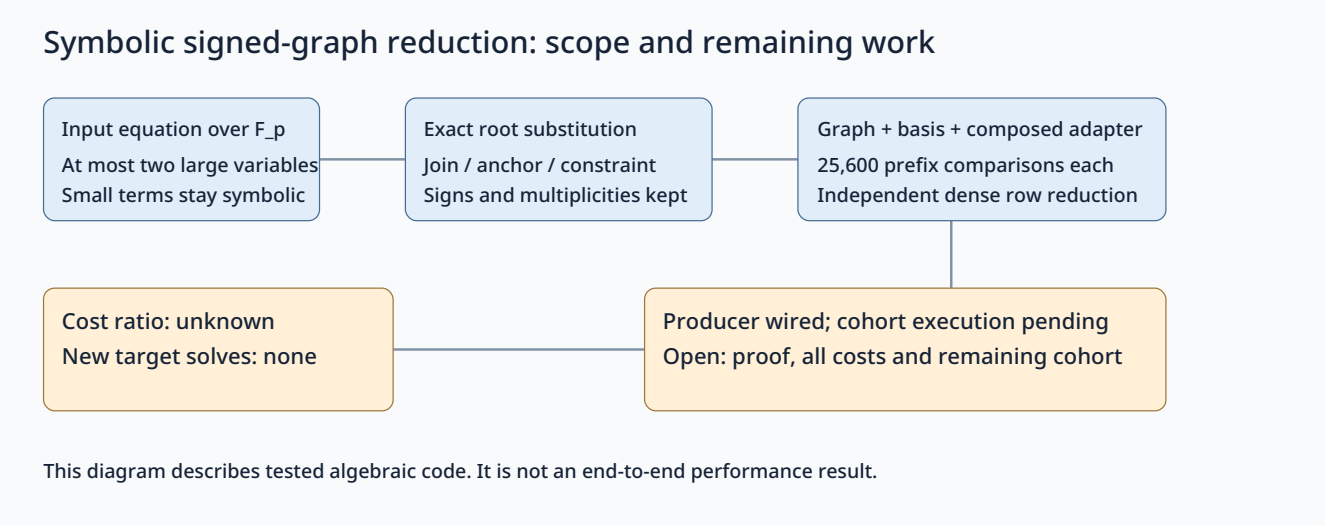
Online timing integration

These are implementation observations for public synthetic mathematical instances, not performance measurements. The root harness uses the existing exclusive phase clock. Target-blind curve, base and empty-solver setup precede target validation. Query/control owns the intervals between named stages; the five exported phase times must sum exactly to the online window. A regression checks this partition and setup exclusion. Additional independent adapter verification remains outside online and is charged in cold group counts. The matched reference opens at target transfer or the first nonzero target multiple. The original interface retains a no-op hook. A regression checks one hook invocation and unchanged answers, operation counts and phase counts in plain, negation and transferred modes. The root zero-coefficient and replay-digest regressions have passed within the full-library run. That run recorded 5,146 passes, seven failures and 123 ignored tests. Six failures required omitted tracked fixtures; the other reported a test-lock conflict. After materialization, all eight selected tests passed in a serial recheck of the same compiled binary. Both logs are retained. The root adapter hashes the deterministic ledger into four exact SHA-256 words for replay. Clock and allocation observations are explicitly excluded; these digest words are identifiers, not charged operation counts. Full native pricing and root allocator observations remain unavailable. Instrumentation overhead is included in clocks. The standalone CLI does not create a phase session and therefore does not export these online times. No qualified host measurement, new cohort outcome or revised ratio is reported here.



Next integration gates

Requirement	Status
Prime-field row kernel and independent prefix tests	Implemented and tested
Forest-state induction and Rust correspondence	Open
Geometric producer wiring	Compile-tested only
Fixed small-base selection and admission accounting	Open
Full symbolic-vector, memory and residual elimination costs	Unmeasured
Remaining 717 frozen candidate targets	No new results
End-to-end cost against matched rho	Unknown

Sources and limitations

The graph construction follows the mathematical distinction between full relations and cycles leaving a large prime in Gaudry, Theriault and Thome, double large prime variation, sections 2.3 and 3.1 (2005 version). Its genus-at-least-three complexity result is not asserted for this genus-2 pencil stream. Existing repository graph and large-prime eliminators informed the implementation. The editable graph is `reduction.svg`; the machine-readable check table is `checks.csv`. All results are development checks; no runtime, memory improvement or full-cohort completion is inferred from them.